



Virtual Streamer Characteristics and Their Impact on Consumer Behavior in Live-Streaming E-Commerce: A Systematic Review

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Abstract: With the advance of live-streaming e-commerce and the metaverse, virtual streamers as a new productive force, are becoming an emerging power in the live-streaming e-commerce industry. The characteristics of virtual streamers and their impact on consumer behavior in live-streaming rooms have gradually attracted academic attention. Although research on virtual streamers is on the rise, there is a lack of integrated synthesis of research findings, especially in the preliminary stage of virtual streamer applications in the e-commerce field. In this light, this paper conducted a holistic review and analysis of the research outcomes related to virtual streamers in the live-streaming e-commerce domain. Firstly, the conceptual connotation and categories of virtual streamers were elucidated. Subsequently, the paper traced back the relevant theories, influencing factors, and research methods concerning virtual streamer characteristics and their impact on consumer behavior. Ultimately, the paper concluded with an outlook for future investigation, in anticipation of promoting advanced application of virtual streamers in e-commerce marketing practices.

Keywords: Virtual streamers; Virtual streamer characteristics; Consumer behavior

1 Introduction

With the rapid development of innovative technologies such as artificial intelligence, generative AI, and virtual reality, new-quality productivity represented by Gen-AI-driven virtual streamers is injecting new vitality into the live-streaming e-commerce industry. In the live-streaming e-commerce context centered on the core elements of "people, products, and scenarios", Gen-AI-driven virtual streamers have achieved innovative breakthroughs in the two dimensions of "people" and "scenarios" by virtue of their unique merits. On the one hand, virtual streamers can provide 24/7 uninterrupted services and conduct personalized interactions with users through generative AI technologies, thus liberating the "people" element. On the other hand, with the support of generative AI, virtual streamers can flexibly adjust the design of virtual live rooms according to the characteristics of different products, thereby generating more attractive visual effects and interactive content. This further brings consumers an immersive experience and reshapes their perception of "scenarios". It was estimated that the market size of industries driven by virtual streamers in China would have reached 640.27 billion yuan by 2025 [1]. During the Double 11 shopping festival in 2025, the total transaction volume of virtual streamers on the platform of JD.com exceeded 2.3 billion yuan, with 17,000 merchants realizing round-the-clock live streaming through digital humans, boosting the platform's overall conversion rate by 30%. This validates the key strength of large-scale commercialization of virtual streamers [2].

Despite the enormous potential and commercial value demonstrated by Gen-AI-driven virtual streamers in live-streaming e-commerce as an emerging form of live commerce, this format is still in its early stage of development. Thus, systematic review of related existing literature is still limited. Triggered by this research gap, this paper first elaborated on the conceptual connotation and classification of virtual streamers. It then reviewed the theoretical foundations of research on how characteristics of virtual streamers influenced consumer behavior. Focusing on Gen-AI-driven virtual streamers in the e-commerce context, this study summarized the influencing factors and research methods adopted in relevant studies. Finally, directions for pioneering research were proposed and discussed. The contributions of this paper are threefold. First, by sorting out the concepts, classification, theoretical foundations, and

research methods of virtual streamers, this study constructed an integrated research framework for comprehending how characteristics of virtual streamers shaped consumer behavior, to lay a solid foundation for subsequent qualitative and quantitative methods. Second, to enrich the theoretical literature on virtual streamers and consumer behavior in e-commerce, this paper summarized the characteristics of virtual streamers and their complex interactions with other factors such as product attributes. It developed a new perspective for understanding how virtual streamers influenced consumer behavior and offered a theoretical basis for future explorations of the underlying impact mechanisms. Third, this review delivered explicit value for intelligent management decision making in live-commerce operations. By synthesizing how characteristics of virtual streamers affect consumer responses, the findings offered actionable insights for managers regarding the design and selection of appropriate types of virtual streamers, the optimization of interaction strategies to enhance consumer engagement, and the matching of streamer characteristics with product categories to improve fit and conversion. The management could also implement consumer segmentation, based on responses to different streamer features, and the refinement of digital marketing and live-commerce operational strategies. In doing so, this review directly bridged academic research and practical decision making; enterprises were then able to leverage Gen-AI-driven virtual streamers more effectively to achieve superior business outcomes.

2 Concepts and Classification of Virtual Streamers

2.1 Concept of Virtual Streamers

The concept of virtual streamers originated from virtual humans mentioned in the 1992 science fiction novel *Snow Crash*, in which real-world individuals possess virtual avatars in the “Metaverse”, a virtual world as termed today. Virtual humans are also referred to as Avatars. In the China Virtual Digital Human Influence Index Report jointly issued by the State Key Laboratory of Media Convergence and Communication at Communication University of China and other institutions, virtual digital humans are defined as interactive virtual figures with human-like appearance, behavior, and thinking, created through technologies including computer graphics, biotechnology, motion capture, brain science, deep learning, and speech synthesis [3]. Some scholars defined virtual humans as technology-created virtual characters driven by humans or Gen-AI, with anthropomorphic appearance and interactive capabilities [4, 5]. On this basis, virtual streamers are expected to possess transaction and functions of social interaction. They can serve as virtual shopping assistants that introduce and promote products in live streaming rooms, while also providing entertainment value beyond shopping through social interactions during live broadcasts [6].

2.2 Classification of Virtual Streamers

Virtual streamers can be classified from multiple perspectives, including their levels of anthropomorphism, application functions, underlying technologies, and visual appearance.

2.2.1 Classification by the level of anthropomorphism

According to the degree of anthropomorphism, virtual streamers can be divided into simple, superficial, intelligent, and digital types [7]. Simple virtual streamers lack human-like appearance and are usually presented as 2D static visuals, performing only low-intelligence tasks. Superficial virtual humans have human-like appearances but limited behavioral capabilities. Intelligent ones exhibit human-like emotions and thinking patterns with relatively low anthropomorphism in appearance. Digital virtual humans combine highly human-like appearances with advanced intelligent emotions, thinking, and perception.

2.2.2 Classification by application functions

Based on application functions, virtual streamers fall into two categories: Service-oriented virtual humans and identity-oriented virtual humans [8]. Service-oriented virtual humans can replace real humans in content production and basic interaction, such as e-commerce virtual streamers, bank consultants, exhibition guides, and virtual hosts, as well as multimodal general AI assistants emphasizing companionship, including health assistants and virtual companions. Identity-oriented virtual humans are closely related to the Metaverse, featuring distinct identity and personality traits, covering various virtual intellectual property (IP) characters, virtual idols, and future virtual avatars representing individuals in virtual world.

2.2.3 Classification by implementation technology

In terms of enabling technology, virtual streamers are categorized into human-driven and AI-driven types [9]. The former uses technologies including text-to-speech (TTS) synthesis, face synthesis, computer graphics (CG) rendering, and motion capture to replicate the expressions, voice, and movements of real people behind the scenes into virtual images, supporting real-time interaction that relies on manual operation. The latter employs 3D modeling and perceptual-cognitive techniques to generate virtual characters and voices, driven by AI to enable 24/7 uninterrupted live streaming [10, 11]. Such AI-driven virtual streamers could respond to complex contexts in real time, while supporting personalized recommendations and generation of intelligent content [12].

2.2.4 Classification by visual appearance

Based on visual image design, virtual streamers are classified into animation, comics, and games (ACG)-style virtual streamers and realistic human-like virtual streamers [13]. ACG-style virtual streamers typically adopt cartoon-based visual designs. Realistic human-like ones use 3D modeling and artificial intelligence to construct virtual streamers that closely resemble real humans in appearance and behavior.

3 Theoretical Foundations for the Impacts of Virtual Streamer Characteristics on Consumer Behavior

The influence of virtual streamer characteristics on consumer behavior is a complex and multi-dimensional process that can be systematically analyzed from perspectives including social behavior, information systems, information processing, and psychology.

3.1 Social Behavior Perspective

Although virtual streamers are non-human entities in nature, individuals often unconsciously regard them as social actors during interactions with consumers. Representative theories mainly include Computers Are Social Actors (CASA) paradigm, anthropomorphism theory, and attachment theory.

3.1.1 Computers Are Social Actors paradigm

CASA paradigm originated in the 1990s and was proposed by Clifford Nass of Stanford University [14]. The core proposition of this theory is that people tend to unconsciously treat computers or other technological products as social actors and exhibit corresponding social responses when interacting with them [15]. Within the CASA framework, even when individuals are fully aware that they are communicating with non-human entities, social cues displayed by virtual streamers such as anthropomorphic appearance, voice, gestures, and facial expressions could act as triggers and activate the same information-processing mechanisms as in interpersonal communication [16]. Therefore, people apply social norms and expectations derived from human-to-human interaction to human-machine interaction, and this tendency is readily influenced by the characteristics of virtual streamers.

3.1.2 Anthropomorphism theory

Anthropomorphism refers to the attribution of human characteristics, motivations, intentions, or emotions to non-human entities, enabling them to exhibit human-like abilities or morphological features [17]. Anthropomorphism is multi-dimensional, encompassing cultural traits, physical appearance, interaction styles, conscious awareness, and emotional expressions, which endow non-human entities with richer human-like attributes [18]. Through anthropomorphism, virtual streamers achieve cross-boundary integration between virtuality and reality and establish social connections between the content of live streaming and consumers. Extant studies have found that the anthropomorphic features of virtual streamers could effectively promote consumers' purchase intention through the mediating role of psychological responses [19–21].

3.1.3 Attachment theory

Attachment is regarded as an innate human response [22]. From academic perspective, attachment is defined as an emotional bond formed between individuals that can persist across time and space [23]. From affective and behavioral perspective, attachment can be categorized into secure, avoidant, and anxious types [24]. Kim and Park introduced attachment theory and argued that the attractiveness of virtual streamers exerted no direct effect on consumers' purchase intention; instead, it influenced consumers' behavioral responses through brand attachment. Meanwhile, the congruence between virtual streamers and products could enhance consumers' purchase intention [25].

3.2 Information Systems Perspective

As an intelligent information system in essence, virtual streamers are notably anthropomorphic in appearance, yet their fundamental nature as technological systems cannot be fully concealed. Accordingly, when interacting with virtual streamers, consumers tend to accept and use them in accordance with the psychological and behavioral patterns applied to information systems. In this process, consumers are also susceptible to information system-related factors such as technology acceptance. Within this perspective, the Technology Acceptance Model (TAM) and the Unified Theory of Acceptance and Use of Technology (UTAUT) have been widely adopted in relevant research. A concrete example can be taken from the study conducted by Luo and Chen, who applied TAM to investigate how virtual idols' live commerce influenced consumers' purchase intention. They revealed that the anthropomorphic characteristics of virtual idols could positively affect consumers' purchase intention of clothing through the serial mediating roles of parasocial interaction, perceived usefulness, and perceived ease of use.

3.3 Psychological Perspective

3.3.1 Uses and Gratifications theory

Conventional information communication theories assume that information senders are typically in an active position. From a psychological perspective, the Uses and Gratifications theory proposed that information senders were indeed influenced and constrained by information receivers [26]. This theory characterizes people's exposure to and use of information communication media such as computers as a process driven by social and psychological factors, which involves forming expectations toward media and achieving a continuous fulfillment of personal needs. It emphasizes understanding consumers' adoption or usage behavior by examining the specific needs and motivations underlying their choice and use of information communication media such as computers.

In research on virtual streamers, individuals watch virtual streamer live broadcasts primarily driven by psychological needs and motivations such as curiosity and the pursuit of novel experiences. During viewing, virtual streamer characteristics including anthropomorphism, interactivity, and professionalism could enhance consumers' immersive experience and generate gratification, thereby promoting their purchase intention [27, 28]. Drawing on the Uses and Gratifications theory, Abdelsattar et al. explored the internal mechanism through which anthropomorphic AI virtual influencers shape consumers' consumption decisions. Their research confirms that emotion-oriented and entertaining content created by virtual influencers fulfills two core psychological demands of audiences, namely emotional companionship and leisure diversion, thereby significantly boosting users' continuous follow intentions and product purchase intentions. This study further verifies that anthropomorphic virtual figures can consistently capture users' attention and steer their consumption behaviors, and audiences' psychological expectations for emotional resonance and group social identification can also be effectively satisfied via virtual content provision [29].

3.3.2 Stimulus-Organism-Response (S-O-R) theory

S-O-R theory provides a core theoretical framework for analyzing the influence of virtual streamer characteristics on consumer behavior from a psychological perspective [30]. This theory explains the internal mechanism by which external stimuli from virtual streamers alter consumers' organic states such as psychological cognition and affective experience, ultimately triggering changes in consumer behavior. Within this theoretical framework, scholars generally regarded the characteristics of virtual streamers as external stimuli faced by consumers in live shopping scenarios. Trust, psychological distance, perceived experiential value, social presence, and similar factors are treated as the psychological perceptions and virtual experiences of the consumers, while purchase intention represents a form of behavioral response to environmental stimuli [31].

3.4 Information Processing Perspective

3.4.1 Elaboration Likelihood Model

The Elaboration Likelihood Model (ELM) is a dual-process information processing model proposed by Petty and Cacioppo in 1986 [32]. This model is widely used to explain how consumers' attitudes toward persuasive information are shaped through two distinct routes: The central route and the peripheral route. The central route emphasizes in-depth processing of the information itself, requiring consumers to carefully evaluate information quality before making judgments. In contrast, the peripheral route focuses on simple cues or peripheral factors, whereby consumers rely on heuristic signals and existing cognition to form decisions.

3.4.2 Heuristic-Systematic Model

The Heuristic-Systematic Model (HSM) of information processing was first proposed by Chaiken and Ledgerwood, to explain how people selected different information processing paths during decision making and judgment [33]. Heuristic processing refers to the mode in which consumers make decisions with minimal cognitive effort, as this relies on simple reasoning. In contrast, systematic information processing requires consumers to channel considerable cognitive effort to carefully evaluate relevant data. In the context of interacting with virtual streamers, consumer responses are initially influenced by social cues that make virtual streamers appear more human like, such as facial expressions, eye contact, gestures, and human-like voices [34]. These features could intuitively and rapidly trigger consumers' heuristic information processing. Meanwhile, the messages delivered by virtual streamers, including details of products and professional knowledge, could stimulate in-depth thinking and thus induce systematic information processing.

4 Research Strategy

4.1 Retrieval from Literature

First, literature search was conducted by using three databases to ensure both disciplinary depth and cross-field comprehensiveness. Specifically, Web of Science served as the foundational database due to their extensive indexing in Social Sciences and Communication research. Google Scholar was adopted for cross-validation and forward/backward citation tracing, while China National Knowledge Infrastructure (CNKI) was included to capture

relevant studies published in Chinese and gain a thorough understanding of domestic research progress in live-streaming e-commerce and virtual streamers. A Boolean search strategy was employed to ensure comprehensive coverage, while maintaining topical relevance. Studies related to virtual streamers and closely associated constructs were identified using the following query: (“virtual streamer” OR “AI streamer*”). This search was intentionally broad to capture the full scope of research on virtual streamers. This syntax was adapted as needed for each database. Given the recent emergence of this topic, no limits of publication date were imposed. This allowed the inclusion of foundational literature, while prioritizing empirical studies published in the past five years. Second, authors independently screened all titles, abstracts, and when necessary, full texts of the articles for eligibility. Discrepancies were discussed and resolved by consensus.

4.2 Screening of Literature

To guarantee the accuracy and objectiveness of the review, all retrieved literature was carefully screened one by one. The inclusion criteria were established as follows: The research topic focused on live-streaming e-commerce or live-streaming sales; the research method included qualitative, quantitative, or mixed methods. Review articles were closely related to virtual streamers and consumer behavior. Articles were initially screened by reading titles and abstracts, and for articles that could not be classified based on preliminary screening, full texts were further reviewed before final judgment.

4.3 Research Content

Ongoing research has paid limited attention to the moderating effects of consumer traits and product types on the influence of virtual streamers. Scholars confirmed that consumers’ diverse external and internal traits could significantly affect the perceived value of products and brands. The kind of products sold also deserved attention: Whether the product is experience-oriented or functional, durable or consumable, will exert varying degrees of influence on consumers’ psychology and behavior in virtual streamer live commerce. Previous studies seldom incorporated virtual streamers, products, and consumers into a unified model, often neglecting the role of one party. Future ground-breaking studies could explore how the interactions among virtual streamer characteristics, product attributes, and traits of consumer groups jointly shape consumer behavior. Specific research questions to be explored are as follows: How should enterprises select AI-driven and human-driven virtual streamers according to product types (experiential vs. functional products; durable goods vs. consumer goods)? How do consumers’ age, gender, and shopping motivation (utilitarian vs. hedonic) affect their trust in virtual streamers and purchase intentions?

Most prior studies explained the impacts of anthropomorphic and interactive characteristics of virtual streamers on consumer behavior, based on anthropomorphism theory and social response theory, while few investigations have been conducted from the perspectives of human-machine relationship and psychological distance. Further follow-up research may draw on AI-related theories in marketing, introduce the concept related to the quality of parasocial relationships, and explore the internal mechanisms through which AI-driven e-commerce virtual streamers influence consumer behavior.

4.4 Research Methods

Most current studies adopted structural equation modeling and experimental approaches to tackle how individual characteristics of virtual streamers (e.g., anthropomorphism, interactivity, and sociability) affect consumer behavior. However, these methods cannot verify the potential interactions and combined effects among multiple attributes. Fuzzy-set qualitative comparative analysis (fsQCA) focused on the complex causal relationships between conditional configurations and outcome variables, thus overcoming the limitation of traditional research that only considers simple linear effects of single variables. Future research could systematically adopt the fsQCA method to analyze how combinations of virtual influencer features (e.g., “high anthropomorphism + high interactivity + low authenticity” or “medium anthropomorphism + high social presence + high responsiveness”) could effectively trigger consumers’ purchase or value co-creation behaviors.

4.5 Scope of Research

Scholars extensively studied virtual streamers in the contexts of news broadcasting, brand endorsement, and virtual idols. However, research on their application in e-commerce marketing is still limited. Compared with other industries, virtual streamers engaging in live-streaming e-commerce require higher levels of interactivity, attractiveness, persuasiveness, and responsiveness to consumer demands. Therefore, future research may deeply investigate the influence of virtual streamers on consumer behavior in e-commerce marketing scenarios, thus providing theoretical support for the multi-scenario application of Gen-AI-driven virtual streamers.

4.5.1 Extended research dimensions of consumer behavior

Previous studies principally focused on the effects of virtual streamers on consumer purchase behavior, while ignoring value co-creation behaviors and citizenship behaviors that e-commerce users exhibit in live rooms, such as commenting, collecting, and sharing. In the era of mobile internet and social media, consumers' value co-creation behaviors (e.g., liking, sharing, and recommending) and citizenship behaviors not only affect purchase intention but also play a crucial role in improving brand reputation, strengthening product identification, and promoting sustainable development of live-streaming commerce. Thus, current research may painstakingly extend the perspective to diversified consumer behaviors, including consumer value co-creation behavior and citizenship behavior.

4.5.2 Comparative differences between human streamers and virtual streamers

Most existing studies focused on the impacts of virtual streamer characteristics on consumer behavior. Although some scholars have examined the differential effects of virtual streamers versus human streamers on consumer behavior, current research essentially adopted between-group experimental designs comparing streamer type and product type, with limited attention to the interactive effects of streamer type with other factors on consumer behavior. In view of this, in-depth studies could be initiated to further identify the differences in the mechanisms through which human streamers and virtual streamers influence consumer behavior under the interaction of other boundary conditions.

As an emerging trend in live-streaming e-commerce, the mechanisms and boundary conditions through which virtual streamer attributes shape consumer behavior urgently require further academic exploration. Pertinent research could draw on AI-related theories in marketing and users' experience, combined with the unique context of live commerce, to profoundly analyze the impacts of virtual streamer characteristics on consumer behavior and the interactive effects among virtual streamers, products, and consumers. This will provide theoretical guidance for the optimal design and marketing application of virtual streamers.

4.6 Research Methods for Investigating the Impacts of Virtual Streamer Characteristics on Consumer Behavior

In studies examining how virtual streamer characteristics influence consumer behavior, scholars employed diverse research approaches. To date, two approaches are dominant: Structural equation modeling and experimental methods.

4.6.1 Structural equation modeling

Based on constructed theoretical models, scholars tested research hypotheses using either covariance-based structural equation modeling (CB-SEM) or partial least squares structural equation modeling (PLS-SEM). CB-SEM estimates model parameters with the covariance matrix of observed variables. Conversely, PLS-SEM is preferred when sample sizes are small or when data violate the normality assumption.

4.6.2 Experimental methods

Experimental methods tested the effects of specific factors on outcomes by controlling relevant variables. Researchers designed distinct experiments to validate how varying levels of virtual streamer attributes shape consumer behavior. Common experimental approaches include eye-tracking experiments and scenario-based experiments.

Eye-tracking experiments: Eye-tracking experiments use eye-tracking devices to record the durations of participants' fixation on different experimental materials, to capture attention-related data. These data were then combined with participants' questionnaire scale ratings for outcome analysis. Chen et al. [20] conducted a 2×2 factorial experiment with two manipulated factors: virtual streamer interactivity (high vs. low) and product type (hedonic vs. utilitarian). Before the main experiment, participants were asked to evaluate the product categories and the perceived interactivity of the virtual streamers while watching live-streaming videos. Finally, regions of interests (ROIs) were defined in the video stimuli, including the virtual streamer area, product area, and interactive bullet-screen area. Using the collected experimental data, the authors verified that virtual streamer interactivity positively influenced consumers' purchase intention.

Scenario-based experiments: Scenario-based experiments involve exposing participants to video clips or live streams under specific contexts, followed by the rating scales in questionnaires. Scholars typically began with single-factor experiments, in which two conditions (high vs. low levels of a virtual streamer characteristic) were set. Group differences between the experimental and control conditions were identified based on participants' video-based ratings. Building on single-factor designs, researchers then introduced a moderating variable with two contrasting levels. Combined with the virtual streamer characteristics, this yielded four experimental conditions. Participants were randomly assigned to these conditions, and their rating scales were applied to test the theoretical model.

5 Impacts of Virtual Streamer Characteristics on Consumer Behavior

5.1 Virtual Streamer Characteristics

Most existing studies focused on appearance-related characteristics that reflected the human-like warmth of virtual streamers, such as anthropomorphism, attractiveness in terms of affability and cuteness [35, 36]. Moreover, e-commerce virtual streamers do not only introduce products and deliver recommendations but also interact with consumers through personalized greetings and other exchanges. Scholars have identified characteristics reflecting streamer competence, including interactivity, professionalism, responsiveness, and sociability [37]. Virtual streamers also exhibited other key traits such as appeal, intelligence, empathy, and entertainment value. The characteristics and their detailed manifestations are presented in Table 1.

Table 1. Characteristics of virtual streamers

References	Characteristics of Virtual Streamers	Description of the Characteristics or Dimensional Segmentation of Virtual Streamers
[31]	Warmth and capability	Warmth: Affection and agility; Ability: Responsiveness
[18]	Anthropomorphism	Interactive personification, representational personification, and cultural personification
[19]	Anthropomorphism	Personification in appearance, morality, cognitive experience, and conscious emotions
[9]	Trustworthiness	Competence, integrity, benevolence
[38]	Interactivity	Interactivity refers to the perceived engagement from real-time, cross-spatiotemporal interaction between consumers and virtual streamers in live-streaming e-commerce
[7]	Sociality	Introduce products, provide recommendations, and share information to guide consumers in completing their purchasing decisions; engage in personalized interactions such as personalized greetings with consumers
[10]	Attractiveness, interactivity, professionalism, and intelligence	Attraction includes appearance, personality, and product suitability; interactivity reflects consumer engagement; professionalism represents in-depth product knowledge and expert recommendation ability; intelligence enables personalized and specialized information delivery based on consumer needs.
[39]	Human-computer interaction quality	Controllability, real-time performance, reliability, professionalism, empathy, and entertainment value

5.2 Impacts of Virtual Streamer Characteristics on Consumer Behavior

Based on a review of relevant literature and theories, virtual streamer characteristics exert multiple effects on consumer behavior. Current research primarily focused on the influences of virtual streamers' traits on consumer purchase behavior, as well as the interactive effects involving consumption motivation, product type, and user characteristics in live-streaming e-commerce. However, studies on consumer citizenship behavior, value co-creation behavior, and their influencing mechanisms in the context of virtual streamer commerce remain limited.

5.2.1 Impacts of virtual streamer characteristics on consumer purchase behavior

In live shopping scenarios, consumers may need to decide between human streamers and virtual streamers. Compared with human streamers, virtual streamers are often more entertaining and appealing and can strengthen viewers' emotional experience in live rooms and further influence purchase intention [40]. Lee et al. noted that the professionalism of virtual streamers enhanced consumers' cognitive trust, thereby improving streamers' persuasiveness and consumer satisfaction with merchants and products, ultimately affecting purchase intention [41]. Characteristics such as personal charisma, entertainment value, and credibility of virtual streaming personas also significantly impact consumers' purchase intention.

The parasocial interaction, attractiveness, perceived realism, and credibility of virtual humans jointly shape consumers' purchase intention [42]. Interactions between virtual streamers and consumers occur across cognitive,

affective, and behavioral dimensions. Interactivity could enhance viewers' sense of engagement and experience, and promote purchase intention by positively affecting consumer engagement, psychological distance, and perceived trust. In addition, interactivity could improve live streaming quality; for instance, virtual reality technology could be applied to deliver a more immersive live streaming experience. Meanwhile, responsiveness also influences consumers' purchase intention.

The design of virtual streamer's external image has also become a research hot spot. Their appearances can take the form of any existing or non-existing living or non-living entities, characterized by anthropomorphism. The anthropomorphic nature of virtual streamers helps build intimate relationships with viewers and reduces psychological distance, thereby influencing consumers' purchase intention [43, 44]. In addition, different communication styles adopted by virtual streamers affect consumers' purchase intention. Some scholars classified the communication styles of virtual streamers into task-oriented and social-oriented types [45]. Yao et al. revealed that social-oriented virtual streamers promoted consumers' purchase intention by enhancing perceived warmth when introducing experience-oriented products, whereas task-oriented virtual streamers enhance purchase intention by demonstrating competence through detailed product information when presenting search products [46].

5.2.2 Interactive impacts of virtual streamer characteristics and other factors on consumer purchase behavior

Scholars have not only focused on the direct impacts of virtual streamer characteristics on consumer purchase behavior but also analyzed the interactive effects between virtual streamer traits and other factors. To date, research has devoted increasing attention to factors such as viewing motivation, product type, and consumer characteristics.

Users' motivations for watching virtual streamer live broadcasts are generally divided into two categories: Hedonic motivation (pursuing hedonic value such as entertainment and relaxation) and utilitarian motivation (pursuing functional value such as information acquisition) [47]. R. J. Wu et al. discovered that differences in the sociability of virtual streamers influenced consumers' experiential value and purchase intention. Specifically, low-sociability task-oriented virtual streamers promoted purchase intention by satisfying consumers' utilitarian motivation, whereas highly social virtual streamers enhanced purchase intention by fulfilling consumers' hedonic motivation [7]. Zhong et al. also observed that the anthropomorphic characteristics of virtual streamers positively affected behavior of consumer citizenship by influencing perceived hedonic value and utilitarian value [18].

Product types differ significantly in their inherent attributes. For products relying on proximal sensory cues (touch, smell, and taste), virtual streamers are less persuasive because they cannot deliver direct experiential sensations. On the other hand, for products dependent on distal sensory cues (hearing, and vision) and objective descriptions, virtual streamers could strengthen persuasive appeal by leveraging their unique advantages. Accordingly, scholars often incorporated product type as a key moderating variable when examining how virtual streamer commerce influenced consumer behavior [48]. Anthropomorphic and other traits of virtual streamers could promote consumer acceptance and purchase intention toward utilitarian products by reducing psychological distance and fostering trust. Meanwhile, interactivity enhanced purchase intention for hedonic products by boosting social presence and improving the shopping experience. Furthermore, researchers have pointed out that virtual streamers were more suitable for promoting search products which required complex information comparison, whereas human streamers performed better in selling experience-oriented products that emphasized affective perceptions [49, 50].

Consumers with various individual traits exhibited varying levels of purchase intention when viewing virtual streamer live commerce. Scholars identified that consumer worship behavior, playful engagement, gender, and age all moderated the effect of virtual streamers on consumers' purchase intention [51, 52].

5.2.3 Impacts on consumer citizenship behavior in live streaming rooms

Consumer citizenship behavior refers to voluntary and altruistic behaviors that go beyond formal consumer roles [53]. In the context of e-commerce live streaming with virtual streamers, consumer citizenship behavior is defined as voluntary actions taken by consumers to improve the quality and effectiveness of virtual live broadcasts. Such behavior interacts with the interactivity of virtual streamers and creates extra value for live streaming beyond traditional shopping experiences. The anthropomorphic characteristics of virtual streamers, as discussed by Zhong et al., exerted significant influences on consumer citizenship behavior, mediated by utilitarian and hedonic shopping value [18].

6 Conclusions

Established literature has provided multi-dimensional theoretical support for understanding how virtual streamer characteristics influence consumer behavior from four perspectives: Social behavior, information systems, psychology, and information processing. These four perspectives complement each other by explaining how virtual streamers affect consumer attitudes and behavioral decisions, respectively, through social interaction rules, technology acceptance logic, psychological perception mechanisms, and information processing pathways. Although the current theoretical foundation is relatively solid, relevant studies tend to apply a single theory independently, thus lacking comparison and integration of the explanatory power of different theories. A unified theoretical framework has not

yet been formed. Potential research could strengthen the integration and cross-validation of multiple theories in the context of live-streaming e-commerce to more thoroughly understand the underlying mechanisms through which virtual streamer characteristics shape consumer behavior.

The available literature has conducted substantial empirical tests on virtual streamer characteristics and highlighted their mechanisms influencing consumer purchase behavior and citizen behavior from the perspectives of key and interaction effects. However, from a holistic perspective, the extant research showed obvious structural imbalances. Anthropomorphism, interactivity, professionalism, and attractiveness are the most frequently investigated characteristics, whereas dimensions such as warmth, sociability, responsiveness, and intelligence remained relatively underexplored. In terms of research findings, the impacts of virtual streamers' appearance style and communication mode on consumer behavior have not reached consistent conclusions, since their effects are easily moderated by boundary conditions such as product type and shopping motivation. These inconsistencies warrant further integration and explanation. Meanwhile, prior studies highly focused on immediate outcomes such as purchase intention yet paid insufficient attention to diverse consumer behaviors including consumer citizenship behavior, value co-creation behavior, sharing behavior, and long-term loyalty. Furthermore, few studies have simultaneously incorporated streamer characteristics, product attributes, and consumer traits to conduct an integrated multi-factor analysis. The research gaps and inconsistent findings further highlight the necessity and value of conducting a systematic review, comparison, and integration of the relevant literature in this paper.

7 Research Summary and Future Research Directions

7.1 Research Summary

By reviewing domestic and international literature on virtual streamers, this paper clarified the conceptual connotation and classification of virtual streamers, and systematically summarized the theoretical foundations, influencing mechanisms, and research methods regarding the impacts of virtual streamer characteristics on consumer behavior. An integrated research framework for the effects of virtual streamer characteristics on consumer behavior is synthesized in this paper (see Figure 1).

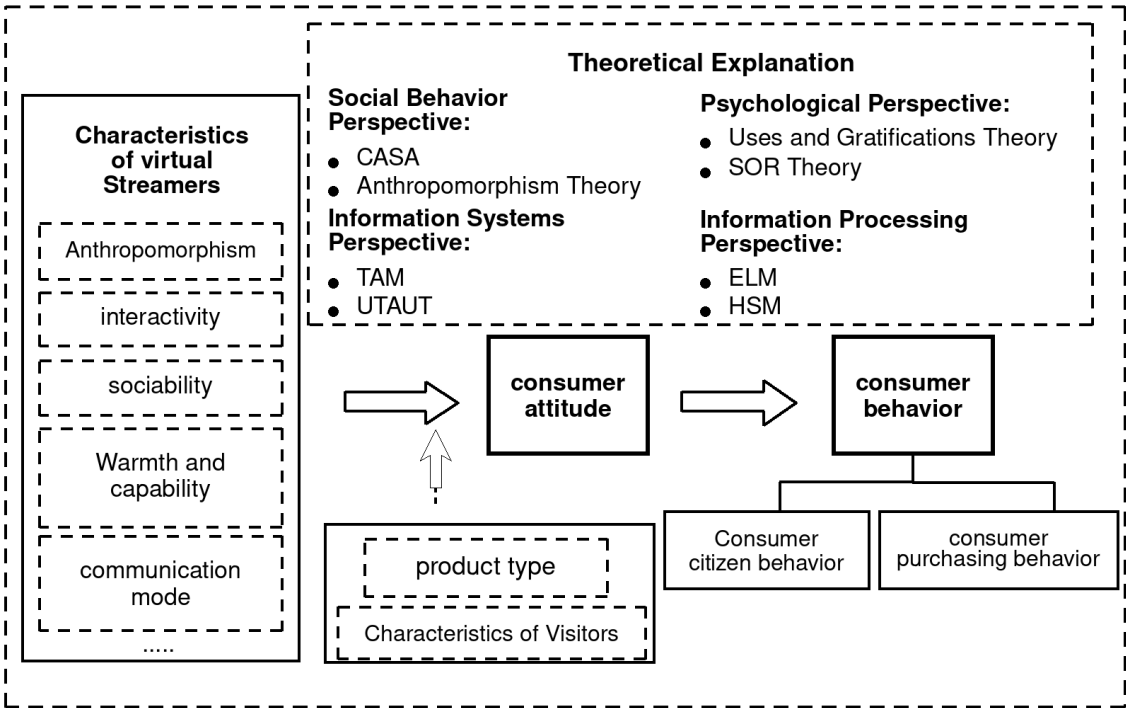


Figure 1. Summary of the study

Note: CASA = Computers Are Social Actors; S-O-R = Stimulus-Organism-Response; TAM = Technology Acceptance Model; UTAUT = Unified Theory of Acceptance and Use of Technology; ELM = Elaboration Likelihood Model; HSM = Heuristic-Systematic Model.

7.2 Managerial Implications

This review provided clear and practical managerial implications for live-streaming e-commerce enterprises. Firms should select virtual streamer types (AI-driven or human-driven, cartoon-style or realistic human-like) based

on operational goals, technical conditions, and target consumer groups. Managers need to design task-oriented interaction for utilitarian consumers and social-oriented interaction for hedonic consumers to improve user experience. Virtual streamers are more suitable for search and utilitarian products, while human streamers perform better in experience products; thus, enterprises should reasonably match product categories with streamer characteristics. In addition, companies can conduct consumer segmentation based on motivation, age, gender, and engagement to achieve precision marketing. They can also optimize live room design, response speed, and content generation by enhancing anthropomorphism, interactivity, and intelligence to reduce psychological distance, strengthen social presence, and ultimately promote purchase intention and consumer citizenship behavior.

7.3 Future Research Directions

Although scholars have conducted numerous exploratory studies on virtual streamers and consumer behavior, previous literature still has several limitations because virtual streamers as a theoretical topic have not been introduced into the field of e-commerce marketing. It is anticipated that intriguing results could be obtained by expanding research in three practical aspects: Multiplying research methods, enriching research content and widening the scope of research.

Author Contributions

Conceptualization, J.L. and J.Z.; methodology, R.Y.; software, J.Z.; validation, J.Z.; formal analysis, J.L.; investigation, J.Z. and R.Y.; resources, J.L. and B.Z.; data curation, J.Z.; writing—original draft preparation, J.Z. and R.Y.; writing—review and editing, J.L.; supervision, J.L. and B.Z.; project administration, J.L. and B.Z.; funding acquisition, J.L. and B.Z. All authors have read and agreed to the published version of the manuscript.

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Data Availability

Not applicable.

Conflicts of Interest

The authors declare no conflicts of interest.

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